

Multiple Linear Regression

Assignment

TTK4260

Scenario: You are a data scientist at *NordicEnergy Analytics*, consulting for the city of Trondheim. The municipality wants to understand and predict building energy consumption to plan sustainable infrastructure. You have been given data from 200 commercial buildings monitored over the past year.

Instructions:

- Solve all questions using a Python notebook with clear explanations.
- Use `plotly` for interactive visualizations where appropriate.
- Show all derivations and intermediate steps for theoretical questions.
- Total points: 150

Question 1 (Multiple Regression Fundamentals – Building Energy Model)

A preliminary analysis focuses on 8 representative buildings. The following data was collected:

Building	Area x_1 (m ²)	Age x_2 (years)	Floors x_3	Energy y (MWh/year)
1	1200	5	3	285
2	2500	15	5	520
3	800	8	2	195
4	3200	25	8	780
5	1800	12	4	390
6	950	3	2	210
7	2100	20	6	545
8	1500	10	4	340

Assume the model: $y = \theta_0 + \theta_1 x_1 + \theta_2 x_2 + \theta_3 x_3 + \varepsilon$

- Set up the design matrix $\mathbf{X}$ (including the intercept column) and response vector $\mathbf{y}$. Write out the matrix dimensions and first three rows explicitly. [5 points]
- Compute $\mathbf{X}^T \mathbf{X}$ and $\mathbf{X}^T \mathbf{y}$ using Python. What is the size of each result? [5 points]
- Solve for $\hat{\boldsymbol{\theta}} = (\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T \mathbf{y}$ and report all four coefficients. [6 points]
- Interpret each coefficient using the “holding others fixed” principle:**
 - What does $\hat{\theta}_1$ tell us about energy consumption per square meter?
 - What does $\hat{\theta}_2$ tell us about older buildings?
 - What does $\hat{\theta}_3$ tell us about adding floors?

Be specific about units and the ceteris paribus assumption.

[8 points]

- Calculate R^2 and the residuals for each building. Which building has the largest prediction error? Suggest a possible explanation. [6 points]

Question 2 (*The Importance of Including Confounders*)

A junior analyst at NordicEnergy initially built a simple model using only building area:

$$y = \theta_0 + \theta_1 \cdot \text{Area} + \varepsilon$$

They found $\hat{\theta}_1 = 0.22 \text{ MWh/m}^2$. However, you suspect this estimate is **biased** because older buildings tend to be larger AND less energy-efficient.

- (a) Using the 8-building dataset from Question 1, fit the simple regression model (Area only) and report $\hat{\theta}_1^{\text{simple}}$. [4 points]
- (b) Now fit the full model including Age. Compare $\hat{\theta}_1^{\text{multiple}}$ from Question 1 with $\hat{\theta}_1^{\text{simple}}$. Which is larger? [4 points]
- (c) **Explain the discrepancy** using the concept of *omitted variable bias*. Draw a simple causal diagram showing how Age relates to both Area and Energy. [6 points]
- (d) Calculate the correlation between Area and Age in the dataset. How does this support your explanation? [4 points]
- (e) A colleague argues: “If our only goal is prediction, we don’t need to worry about confounders.” Do you agree? Explain when this statement is true and when it fails. [6 points]

Hint: Consider what happens if the correlation structure changes in future data.

Question 3 (*Multicollinearity and Coefficient Stability*)

NordicEnergy’s full dataset includes a feature called “Total Window Area” (x_4) which is highly correlated with building Area (x_1). A sample of 10 buildings shows:

ID	x_1 (Area, m ²)	x_4 (Windows, m ²)	x_2 (Age)	y (Energy)
1	1000	180	10	245
2	1500	275	8	358
3	2000	360	15	485
4	1200	220	5	278
5	2500	455	20	615
6	1800	330	12	425
7	900	165	7	218
8	2200	400	18	538
9	1600	290	11	382
10	1100	200	6	262

- (a) Calculate the Pearson correlation coefficient between x_1 and x_4 . What does this suggest about including both in a regression? [4 points]
- (b) Fit **three models** and report the coefficients:
 - Model A: $y = \theta_0 + \theta_1 x_1 + \theta_2 x_2 + \varepsilon$
 - Model B: $y = \theta_0 + \theta_4 x_4 + \theta_2 x_2 + \varepsilon$
 - Model C: $y = \theta_0 + \theta_1 x_1 + \theta_4 x_4 + \theta_2 x_2 + \varepsilon$

[8 points]

- (c) Compare R^2 for all three models. Does including both x_1 and x_4 substantially improve the fit? Explain. [5 points]

- (d) **Bootstrap analysis:** Resample the data 500 times (with replacement) and fit Model C each time. Plot histograms of $\hat{\theta}_1$ and $\hat{\theta}_4$ across bootstrap samples. What do you observe about the variance? [8 points]

- (e) Calculate the Variance Inflation Factor (VIF) for each predictor in Model C:

$$\text{VIF}_j = \frac{1}{1 - R_j^2}$$

where R_j^2 is the R^2 from regressing x_j on the other predictors. Interpret the results. [7 points]

- (f) **The “Who Gets Credit” Problem:** For Model C, show that many different combinations of (θ_1, θ_4) give nearly identical predictions. Illustrate this graphically by plotting the constraint $\theta_1 \cdot \bar{x}_1 + \theta_4 \cdot \bar{x}_4 \approx \text{constant}$. [8 points]

Question 4 (Categorical Variables and Dummy Encoding)

The building dataset includes a categorical variable: **Building Type** with three categories: *Office*, *Retail*, and *Industrial*. Here is an extended sample:

ID	Area (m ²)	Type	Age (years)	Energy (MWh)
1	1500	Office	10	380
2	1200	Retail	5	420
3	2000	Industrial	15	520
4	1800	Office	8	440
5	1000	Retail	3	355
6	2500	Industrial	20	640
7	1600	Office	12	395
8	1400	Retail	7	485
9	2200	Industrial	18	570
10	1100	Office	6	285
11	1300	Retail	4	450
12	1900	Industrial	12	495

- (a) **The Dummy Variable Trap:** Explain why we cannot include dummy variables for all three building types plus an intercept. What mathematical problem would this cause? [5 points]
- (b) Create dummy variables D_{Retail} and $D_{\text{Industrial}}$, using *Office* as the reference category. Show the encoding for the first 6 buildings. [5 points]
- (c) Fit the model:

$$y = \theta_0 + \theta_1 \cdot \text{Area} + \theta_2 \cdot \text{Age} + \theta_R \cdot D_{\text{Retail}} + \theta_I \cdot D_{\text{Industrial}} + \varepsilon$$

Report all coefficients. [6 points]

- (d) **Interpret the categorical coefficients:**

- What is the expected energy consumption for an Office building?
- How much more/less energy does a Retail building use compared to Office?
- How much more/less energy does an Industrial building use compared to Office?

[8 points]

- (e) Your manager asks: “What is the difference in energy consumption between Retail and Industrial buildings?” Calculate this and explain how you obtained it from the model. [4 points]

Question 5 (*Feature Scaling and Its Importance*)

The full NordicEnergy dataset has features with vastly different scales:

Feature	Range	Units
Building Area	500 – 5000	m ²
Age	1 – 50	years
Average Temperature	-5 – 25	°C
Occupancy	10 – 500	people
Electricity Price	0.8 – 1.5	NOK/kWh

Consider the following sample data (10 buildings):

ID	Area	Age	Temp	Occupancy	Price	Energy
1	1500	10	8	120	1.1	385
2	3000	25	5	250	0.9	720
3	800	5	12	50	1.3	195
4	2200	15	7	180	1.0	525
5	4000	35	3	350	0.85	945
6	1200	8	10	80	1.2	295
7	2800	20	6	220	0.95	665
8	1000	3	15	70	1.4	255
9	3500	30	4	300	0.88	830
10	1800	12	9	140	1.05	435

- (a) Fit an OLS model using **unscaled** features. Report all coefficients with their units. [5 points]
- (b) Can you directly compare the coefficient magnitudes to determine which feature is “most important”? Explain why or why not. [4 points]
- (c) Apply **Z-score standardization** to all features:

$$x_j^* = \frac{x_j - \mu_j}{\sigma_j}$$

Fit the model on standardized features and report the *standardized coefficients*. [6 points]

- (d) Now which feature has the largest effect (in terms of standard deviations)? Interpret the largest standardized coefficient. [5 points]
- (e) **Gradient Descent Comparison:** Implement gradient descent with learning rate $\alpha = 0.01$ for:
- Unscaled features
 - Standardized features

Plot the cost function over 1000 iterations for both. What do you observe? [10 points]

- (f) **Critical Rule:** Explain why you must “fit the scaler on training data only, then apply to test data.” What could go wrong if you scale using test data statistics? [5 points]

Question 6 (Regularization: Ridge, Lasso, and Feature Selection)

NordicEnergy has collected a high-dimensional dataset with 30 features for 100 buildings. However, the engineering team believes only about 5–8 features truly matter for energy prediction. A synthetic version is provided below (you will generate the full data programmatically).

Data Generation (use this code):

```
1 import numpy as np
2 np.random.seed(42)
3 n, p = 100, 30
4 X = np.random.randn(n, p)
5 # True coefficients: only first 5 are non-zero
6 true_beta = np.zeros(p)
7 true_beta[:5] = [8, -5, 3, -2, 4]
8 y = X @ true_beta + np.random.randn(n) * 2
```

- (a) Fit an OLS model and examine the coefficients. How many coefficients are “close to zero” (say, $|\hat{\theta}_j| < 0.5$)? How does this compare to the truth? [5 points]
- (b) Split the data into training (70%) and test (30%) sets. Calculate training and test MSE for OLS. Is there evidence of overfitting? [5 points]

- (c) **Ridge Regression (L2):** Implement Ridge regression with the objective:

$$\min_{\boldsymbol{\theta}} \|\mathbf{y} - \mathbf{X}\boldsymbol{\theta}\|^2 + \lambda \|\boldsymbol{\theta}\|_2^2$$

Fit models for $\lambda \in \{0.001, 0.01, 0.1, 1, 10, 100\}$ and plot the *regularization path* (coefficients vs λ). [8 points]

- (d) **Lasso Regression (L1):** Implement Lasso regression:

$$\min_{\boldsymbol{\theta}} \|\mathbf{y} - \mathbf{X}\boldsymbol{\theta}\|^2 + \lambda \|\boldsymbol{\theta}\|_1$$

Fit models for the same λ values and plot the regularization path. [8 points]

- (e) **Compare the paths:** At what λ does Ridge first shrink all coefficients below 1? At what λ does Lasso first set any coefficient to exactly zero? What explains this difference geometrically? [8 points]
- (f) **Optimal λ Selection:** Use 5-fold cross-validation to find the optimal λ for both Ridge and Lasso. Report the best λ and corresponding test MSE for each method. [8 points]
- (g) For Lasso at the optimal λ : How many coefficients are exactly zero? Which features were selected? Compare with the true non-zero features. [6 points]
- (h) **Geometric Visualization:** Create a 2D plot showing:

- The L2 constraint region (circle): $\theta_1^2 + \theta_2^2 \leq C$
- The L1 constraint region (diamond): $|\theta_1| + |\theta_2| \leq C$
- Elliptical OLS loss contours
- The OLS, Ridge, and Lasso solutions

Explain why Lasso solutions tend to lie on corners (axes). [10 points]

- (i) **The Correlated Predictor Caveat:** Modify the data generation so that $x_6 = x_1 +$ small noise. Fit Lasso again. What happens to the coefficients $\hat{\theta}_1$ and $\hat{\theta}_6$? Is this stable across different random seeds? Discuss the implications for interpretation. [7 points]

Question 7 (Comprehensive Application: Trondheim Building Energy Model)

NordicEnergy wants you to build a production-ready model for Trondheim's building energy consumption. You will use a larger synthetic dataset that mimics real patterns.

Data Generation:

```
1 import numpy as np
2 import pandas as pd
3 np.random.seed(2024)
4 n = 200
5
6 # Generate features
7 area = np.random.uniform(500, 5000, n)
8 age = np.random.uniform(1, 50, n)
9 floors = np.random.randint(1, 15, n)
10 building_type = np.random.choice(['Office', 'Retail', 'Industrial'], n,
11                                  p=[0.5, 0.3, 0.2])
12 # Energy depends on features + type effects + interactions
13 base_energy = 50 + 0.15*area + 2*age + 15*floors
14 type_effect = np.where(building_type=='Office', 0,
15                         np.where(building_type=='Retail', 80, 50))
16 energy = base_energy + type_effect + np.random.normal(0, 50, n)
17
18 df = pd.DataFrame({'Area': area, 'Age': age, 'Floors': floors,
19                   'Type': building_type, 'Energy': energy})
```

(a) Exploratory Data Analysis:

- Create pairwise scatter plots of all numeric features vs Energy
- Calculate and visualize the correlation matrix
- Show boxplots of Energy by Building Type

Summarize your key findings.

[10 points]

(b) Model Building: Fit the following models and report R^2 and RMSE for each:

- Model 1: Energy $\sim$ Area (simple regression)
- Model 2: Energy $\sim$ Area + Age + Floors
- Model 3: Energy $\sim$ Area + Age + Floors + Type (with dummies)
- Model 4: Model 3 + interaction (Type $\times$ Age)

[12 points]

(c) Feature Scaling & Regularization:

- Standardize all numeric features
- Apply Ridge and Lasso regression with cross-validated λ
- Compare with OLS in terms of test set performance

[10 points]

(d) Model Diagnostics: For your best model:

- Plot residuals vs fitted values
- Create a Q-Q plot of residuals
- Test for heteroscedasticity

- Identify any influential observations

[10 points]

(e) **Interpretation Report:** Write a 200-word executive summary for Trondheim municipality explaining:

- Which factors most influence building energy consumption?
- How do different building types compare?
- What actionable recommendations can you provide?

Use appropriate uncertainty quantification (confidence intervals).

[10 points]

(f) **Prediction Tool:** Create a function that takes building characteristics as input and returns:

- Point prediction for energy consumption
- 95% prediction interval

Test it on a hypothetical new 2000 m² Office building that is 10 years old with 5 floors.

[8 points]

Question 8 (Bonus: Elastic Net and Advanced Regularization)

This question is optional and worth extra credit.

The Elastic Net combines L1 and L2 penalties:

$$\min_{\boldsymbol{\theta}} \|\mathbf{y} - \mathbf{X}\boldsymbol{\theta}\|^2 + \lambda_1 \|\boldsymbol{\theta}\|_1 + \lambda_2 \|\boldsymbol{\theta}\|_2^2$$

- Explain the geometric constraint region for Elastic Net. How does it differ from pure L1 or L2? Visualize it for 2 coefficients. [5 bonus points]
- Using the data from Question 6 (with correlated x_1 and x_6), fit Elastic Net with various combinations of λ_1 and λ_2 . Does it handle the correlated predictors better than pure Lasso? [5 bonus points]
- Research and implement the “grouping effect” property of Elastic Net. Show empirically that correlated predictors tend to have similar coefficients. [5 bonus points]